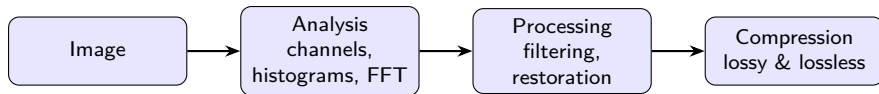


An Image Processing Workspace

Signals, Transforms, and Compression: Concepts and Algorithms

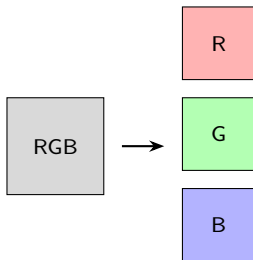
The Big Picture

- An image is a 2D signal: analyze it, transform it, compact it.
- Every tool here is one of those three moves.



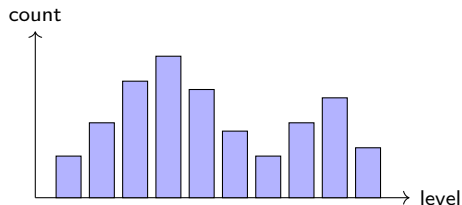
Channel Separation

- RGB = three stacked signals $R(x, y)$, $G(x, y)$, $B(x, y)$.
- BGR vs. RGB load order is a classic bug, caught by viewing channels.



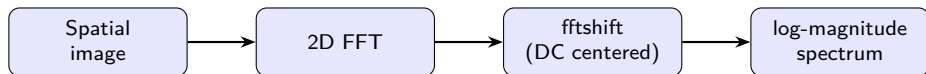
Histograms: Statistics of a Signal

- Histogram = empirical distribution of pixel intensities.
- Luminance blends channels *per pixel first*: $Y = 0.299R + 0.587G + 0.114B$.



Frequency Content: The 2D FFT

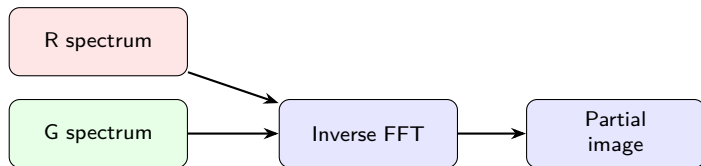
- Image $\rightarrow$ sum of 2D sinusoids of varying spatial frequency.
- `fftshift` centers DC; log-magnitude makes the spectrum viewable.



Parseval: energy in space = energy in frequency

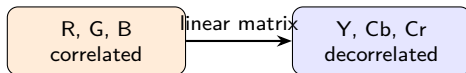
Partial Reconstruction

- Inverse FFT is linear: rebuild from a subset of channel spectra.
- *Phase*, not magnitude, carries most of the structure.



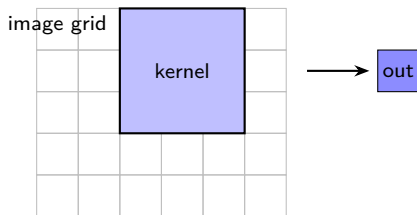
Color Space: RGB vs. YCbCr

- Linear change of basis: decorrelates brightness from color.
- Y carries detail; C_b, C_r are smooth $\Rightarrow$ energy compaction.



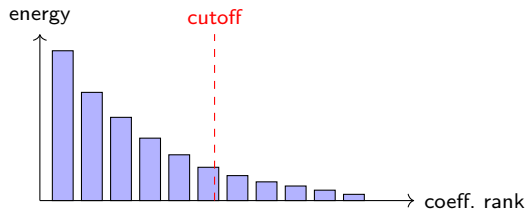
Spatial Filtering: Convolution

- A kernel slides over the image; each pixel becomes a weighted local sum.
- Low-pass = blur (box, Gaussian); high-pass = sharpen (Laplacian, unsharp).



Lossy Compression: Fourier Energy Pruning

- Rank frequency coefficients by energy; keep a target energy %.
- Real images: conjugate-symmetric spectrum $\Rightarrow$ store half.



Lossless Compression: Integer Wavelet Lifting

- Integer Haar wavelets separate the image into averages and details.
- Keep every coefficient and compress the resulting patterns.
- Reverse the steps to recover the exact RGB image.

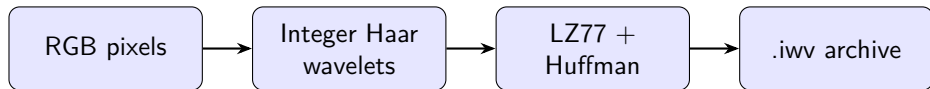


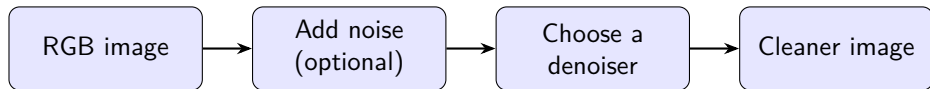
Photo Restoration

- White top-hat subtracts the opened grayscale image to reveal bright marks.
- A repair mask selects the damaged areas.
- Inpainting estimates replacement colors from nearby pixels.



Noise: Models and Denoising

- Gaussian noise adds random variation; salt-and-pepper adds black or white dots.
- Median, Gaussian blur, or non-local means reduce noise.
- Denoising estimates a cleaner image and may soften some details.



Recurring Threads

